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Study of Various Network Commands Used in Linux and Windows

AIM:

To study and execute various network commands used in Linux and Windows for checking network connectivity, viewing IP configuration, identifying network routes, resolving domain names, and monitoring network connections.

ALGORITHM:

1. Start the computer and open Command Prompt in Windows or Terminal in Linux.
2. Execute the hostname command to display the computer name.
3. Execute ipconfig /all in Windows or ip address show in Linux to view network configuration details.
4. Execute arp -a to display the IP and MAC address mappings.
5. Execute ping google.com to check connectivity to a remote host.
6. Execute tracert google.com in Windows or traceroute google.com in Linux to identify the route taken by packets.
7. Execute netstat -an to display active network connections and listening ports.
8. Execute nslookup google.com to obtain DNS information.
9. Execute route print in Windows or ip route show in Linux to display the routing table.
10. Observe the outputs and record the results of each command.

COMMANDS AND OUTPUT:

1. HOSTNAME

Command:
hostname

Output:
DESKTOP-ABC123

Result: Displays the name of the computer.

2. IP CONFIGURATION

Command:
ipconfig /all

Output:
Ethernet adapter Ethernet:
IPv4 Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Result: Displays the IP address, subnet mask, default gateway, and other network configuration details.

3. ARP COMMAND

Command:
arp -a

Output:
Internet Address Physical Address
192.168.1.1 AA-BB-CC-DD-EE-FF

Result: Displays the IP addresses and their corresponding MAC addresses stored in the ARP table.

4. PING COMMAND

Command:
ping google.com

Output:
Reply from 142.250.xxx.xxx: Bytes=32 time=20ms TTL=115

Result: Checks the connectivity between the local computer and the destination host.

5. TRACERT COMMAND

Command:
tracert google.com

Output:
Tracing route to google.com
1 2 ms 3 ms 2 ms 192.168.1.1
2 10 ms 12 ms 11 ms ISP Gateway

Result: Displays the intermediate network devices through which packets travel to reach the destination. The output shown is illustrative.

6. NETSTAT COMMAND

Command:
netstat -an

Output:
Proto Local Address Foreign Address State
TCP 192.168.1.10:5000 192.168.1.20:80 ESTABLISHED

Result: Displays active network connections, listening ports, and connection states.

7. NSLOOKUP COMMAND

Command:
nslookup google.com

Output:
Server: dns-server
Address: 192.168.1.1

Name: google.com
Address: 142.250.xxx.xxx

Result: Resolves the domain name into its corresponding IP address using DNS.

8. ROUTING TABLE COMMAND

Command:
route print

Output:
Network Destination Netmask
0.0.0.0 0.0.0.0
192.168.1.0 255.255.255.0

Result: Displays the routing table used by the computer to determine the path for sending network packets.

RESULT:

Thus, various network commands used in Linux and Windows were studied and executed. The commands were used to examine IP configuration, test network connectivity, identify network routes, resolve domain names, view ARP mappings, and monitor active network connections.